

2946. Matrix Similarity After Cyclic Shifts

Difficulty: Easy

Link: <https://leetcode.com/problems/matrix-similarity-after-cyclic-shifts/description/?envType=daily-question&envId=2026-03-27>

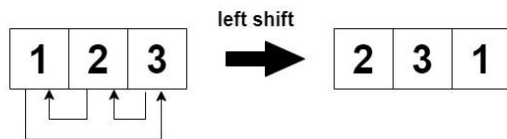
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Problem:

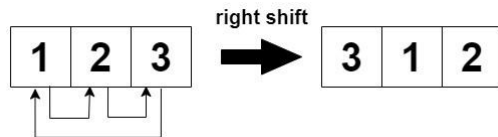
You are given an $m \times n$ integer matrix `mat` and an integer `k`. The matrix rows are 0-indexed.

The following process happens `k` times:

- **Even-indexed** rows (0, 2, 4, ...) are cyclically shifted to the left.



- **Odd-indexed** rows (1, 3, 5, ...) are cyclically shifted to the right.



Return `true` if the final modified matrix after `k` steps is identical to the original matrix, and `false` otherwise.

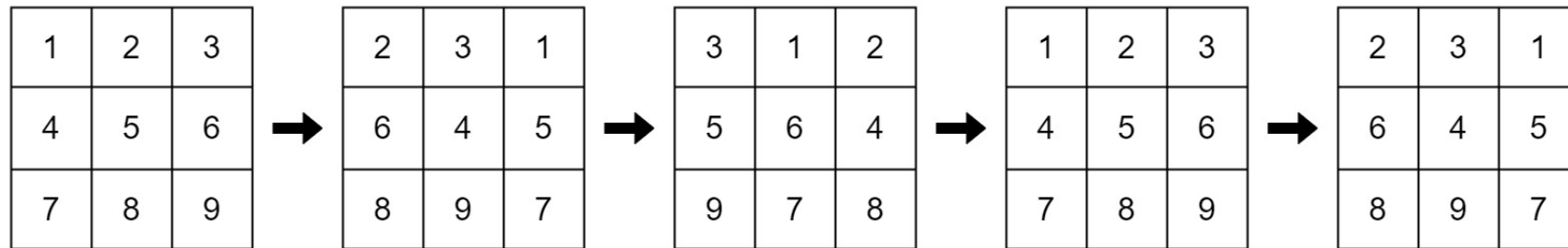
Example 1:

Input: mat = [[1,2,3],[4,5,6],[7,8,9]], k = 4

Output: false

Explanation:

In each step left shift is applied to rows 0 and 2 (even indices), and right shift to row 1 (odd index).

**Example 2:**

Input: mat = [[1,2,1,2],[5,5,5,5],[6,3,6,3]], k = 2

Output: true

Explanation:



Example 3:

Input: mat = [[2,2],[2,2]], k = 3

Output: true

Explanation:

As all the values are equal in the matrix, even after performing cyclic shifts the matrix will remain the same.

Constraints:

- `1 <= mat.length <= 25`
- `1 <= mat[i].length <= 25`
- `1 <= mat[i][j] <= 25`
- `1 <= k <= 50`